

Study of ultra-fine Redistribution Layers based on Hybrid Glass Substrate[†]

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Abstract

This paper presents a hybrid structure integrating ultra-fine redistribution layers (RDLs) with glass substrate. RDLs with 2/2 line/space features were achieved using an embedded process, providing a wiring density 2.5 times that of conventional substrates. The structure eliminates the solder interface, reducing the package profile and lamination steps, thereby improving cost and yield. In reliability tests, the structure showed $\leq 5\%$ resistance variation without peeling or void formation, supporting AI/HPC application.

Author Keywords

ultra-fine RDLs, hybrid glass substrate, fine-line layer

1. Introduction

In the post-Moore era, as transistor dimensions approach physical and lithographic limits, advanced packaging technology has become a central pathway for continued chip performance enhancement [1-3]. In artificial intelligence and high-performance computing, the increasing demand for high-density wiring, large-form-factor packages, and low transmission loss is accelerating the development of advanced packaging technologies such as 2.5D and 3D integration. The core structure of 2.5D and 3D packaging involves organic substrates and interposer.

On one hand, traditional organic substrates, limited by poor antiwarpage capability and insufficient dielectric properties, are increasingly unable to meet the requirements of multichiplet integration and large-form-factor packaging. In addition, constrained by the filler particle size in organic build-up materials, traditional laser-drilling processes face challenges in achieving high-density interconnects (vias $< 10\ \mu\text{m}$) [4]. On the other hand, although silicon interposers provide high wiring precision enabling die-to-die interconnection, they suffer from high cost, thermal mismatch, and large package thickness. Packaging configurations using interposers also require at least two assembly and test cycles, resulting in elevated costs. To address these challenges, this study proposes an asymmetric packaging structure that monolithically integrates an organic interposer with a glass substrate and achieves ultra-fine RDL fabrication through process innovation, offering a promising solution for next-generation advanced packaging. The glass substrate, with its excellent dimensional stability, superior electrical performance (e.g., lower dielectric loss and reduced signal crosstalk), and stronger antiwarpage capability, is considered a key material for overcoming current packaging bottlenecks. The organic interposer provides high-precision wiring ($L/S \leq 2/2\ \mu\text{m}$), enhancing wiring density and enabling one-time assembly of chiplets and the substrate.

This paper aims to develop a prototype of a high-density RDL on hybrid glass substrate. The prototype adopts an asymmetric stacked structure of “fine-line layer (FLL) + glass core.” Its wiring layer configuration can be represented as $X+(2/2/3)$, where X denotes 1–3 layers of fine-line wiring from the organic

interposer. To clearly describe the core process and performance, this study focuses on the example of a single FLL to demonstrate the achievement of ultra-fine L/S and to systematically present and discuss the corresponding reliability test results.

2. Experiment

To meet the packaging requirements of high-bandwidth memory and wide I/O interfaces, an integrated FLLs prototype based on a glass core board was designed, as shown in Figure 1. The prototype adopts an asymmetric structure: the glass substrate side contains two metal core layers, three ABF build-up layers, a solder mask, and a ball grid array on the lower side; the upper side of the core contains two ABF build-up layers and the FLL, where the FLL section includes X layers of high-density wiring. The overall configuration is $X+2/2/3$ ($X=1-3$). The glass core thickness is 0.7 mm, and the thickness of a single FLL is 5–10 μm . To comprehensively evaluate performance, five L/S test patterns were designed within the FLL: 2/2 μm , 2.5/2.5 μm , 3/3 μm , 4/4 μm , and 5/5 μm , along with microvias of two diameters (10 μm and 20 μm). The metal wiring thickness was set at 2–3 μm , pad pitch at 40 μm , and copper pad diameter at 25 μm . These dimensions were selected to ensure efficient pitch transition between the chip and the glass substrate.

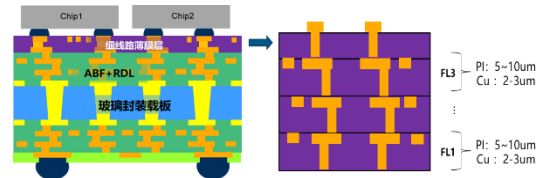


Figure 1. Schematic of the ultra-fine RDLs based on Hybrid Glass Substrate

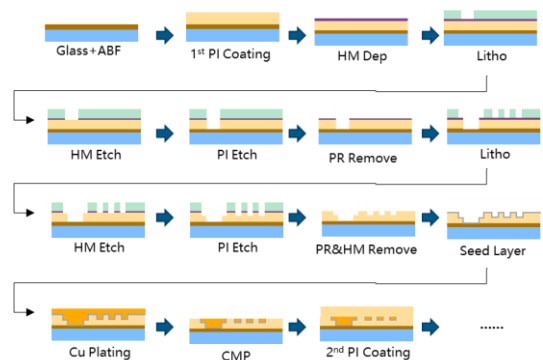


Figure 2. Process Flow of FLLs

The FLL was fabricated using an embedded wiring process, with the sequence shown in Figure 2: ① After pre-treating the ABF substrate, polyimide (PI) insulating material was spin-coated and cured at 180 $^{\circ}\text{C}$ for 60 min; ② A hard mask (HM) material was deposited, patterned via lithography, and selectively etched to

remove the HM in the window regions; ③ Dry etching was used to remove the PI not protected by the HM, forming microvia and microtrench structures, followed by removal of the remaining HM; ④ A titanium/copper seed layer was deposited by magnetron sputtering, after which copper was electroplated to the target thickness; ⑤ Chemical mechanical polishing (CMP) was performed to remove excess surface metal, forming a planarized single FLL. By repeating these steps, multilayer FLL structures can be constructed. Key process optimization considerations include HM material selection, thickness matching between the photoresist and HM, and control of the gas composition and dry-etching parameters.

3. Results and Discussion

In traditional organic substrate packaging, the independent use of ABF and PI build-up layers has been well established. However, the adhesion and reliability between ABF and PI in hybrid substrate applications have not been verified. Therefore, evaluating the bonding strength between the first FLL and the underlying ABF dielectric layer is critical for ensuring the reliability of the hybrid structure. A test plan was designed to assess the adhesion performance of various PI materials on the ABF surface. In this test method, PI was spin-coated and cured on a pre-treated ABF substrate, bonded to a test stub using epoxy resin, and subjected to a mechanical pull test until interface failure. The failure strength and failure mode (i.e., whether failure occurred at the ABF/PI interface) were recorded. As shown in Table 1, the average adhesion strength of six tested PI materials exceeded 70.1 MPa. For samples 1–5, failure occurred between the PI and epoxy rather than at the ABF/PI interface, indicating that the ABF/PI bond strength surpassed the cohesive strength of the test epoxy (~70 MPa). Sample 6 failed at the glass/ABF interface, which was attributed to improper ABF lamination or curing parameters that caused slight surface wrinkling and reduced adhesion to the glass (55.8 MPa). After subsequent reliability tests, SEM inspection of all sample edges revealed no peeling or void between ABF and PI. The pull test and reliability test results collectively demonstrate excellent interfacial adhesion between the PI material and ABF, supporting the feasibility of developing the hybrid glass substrate integrated with FLLs.

Table 1. PI Adhesion Tests

PI	Adhesion /Mpa	Broken interface
1	69.8	PI/Epoxy
2	70.7	PI/Epoxy
3	71..3	PI/Epoxy
4	69.5	PI/Epoxy
5	69.3	PI/Epoxy
6	55.8	ABF/Glass

Traditional organic substrates typically use a semi-additive process (SAP) for fine-line wiring, which often requires a thick plating photoresist as a mask. However, excessively thick photoresist affects light scattering and chemical uniformity during exposure and development, making it difficult to achieve fine RDL lines and spaces ($\leq 4/4 \mu\text{m}$). In addition, SAP-formed lines can collapse during subsequent photoresist stripping. To address these issues, this work proposes a process route combining a hard mask with embedded wiring: a thin HM layer is first patterned using relatively thin photoresist, then high-

precision patterns are formed in the PI via dry etching under HM protection. The etching process employs a mixed gas composed of etch A and B, each with high selectivity for PI and HM materials, respectively. The key parameter governing this scheme is the Etch Rate Ratio (ERR), defined as the ratio of the etch rate of the HM material to the etch rate of the PI under identical conditions. A lower ERR indicates minimal HM consumption during PI etching, resulting in more consistent window profile. An ERR greater than 1 may lead to complete HM removal before the PI is fully etched. This study compared the ERR characteristics of two inorganic materials and one organic material, with results summarized in Table 2. Materials 1 and 2 had ERR values close to 1, indicating HM consumption rates similar to PI, which would not provide stable etching performance. These materials would also require much higher HM thickness compared to PI, causing issues such as poor pattern consistency and increased etching costs. Material 3 exhibited a very low ERR value of 0.057, demonstrating excellent etch-stop capability: only $0.57 \mu\text{m}$ HM was required to etch $10 \mu\text{m}$ PI. Based on these results, Inorganic-2 was selected as the HM material for subsequent PI etching studies.

Table 2. Etch Rate Ratio of Different HM Materials

HM	ERR
Organic-1	0.97
Inorganic-1	1.02
Inorganic-2	0.057

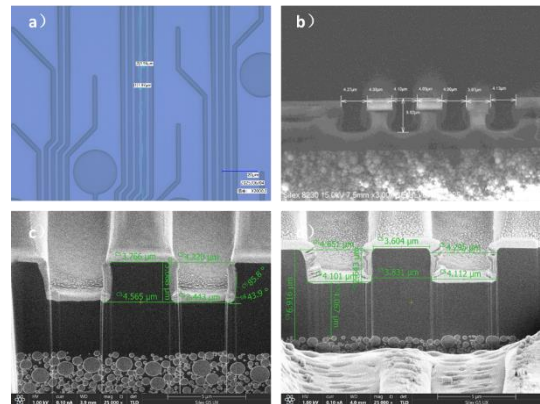


Figure 3. SEM images of Microtrench

a) Optical image of $2/2 \mu\text{m}$; b) Cross-section under low proportion of etch A; c) Cross-section after etch ratio optimization; d) Cross-section under optimal conditions.

As shown in Figure 2, the etching process consists of two steps—HM etching and PI etching—each requiring different primary etch atmospheres. Therefore, the etch gas ratio is a key factor affecting microtrench morphology. Figure 3 illustrates the resulting trench shapes under various etching conditions. After process optimization, Figure 3a demonstrates that microtrenches with $L/S = 2/2 \mu\text{m}$ can be stably achieved with clear edges and no serrated morphology. When the proportion of etch A was low, gas exchange and reactant replenishment at the trench center were better than at the edges, resulting in faster etch rates along both sides and forming pronounced side grooves (Figure 3b).

Increasing the proportion of etch A significantly reduced the side grooves (Figure 3c). With further optimization, when the etch A proportion reached the optimal condition, the side grooves were essentially eliminated, yielding an etch depth of 2.6 μm and a sidewall verticality greater than 85° (Figure 3d). Only slight tailing appeared at the trench bottom corners, caused by small rate differences between the center and edges near the end of etching; this effect is not expected to significantly impact RDL electrical performance.

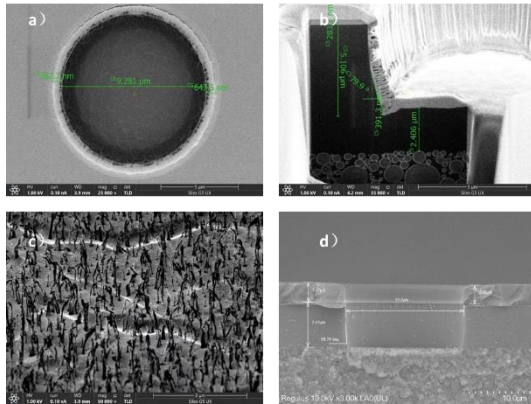


Figure 4. SEM images of Microvia

- a) 10 μm via; b) Cross-section after etching @5 μm ; c) Polymer accumulation on HM surface; d) Cross-section after optimized etching @7.5 μm .

Photosensitive polyimide (PSPI) enables direct patterning through exposure and development; however, the resolution of mainstream commercial PSPI is typically limited to 10 μm , and the difference between top and bottom via diameters is often large, thereby constraining via density. Dry etching can overcome this limitation, enabling smaller vias with steeper sidewalls and more uniform top/bottom dimensions, which in turn enhances achievable interconnect density. In this work, the target microvia depth was 7.5 μm . As shown in Figure 4a, the optimized dry-etching process produced well-formed, residue-free 10 μm microvias, which is difficult to obtain through traditional ABF substrate laser-drilling methods. The study found that under conditions with a high proportion of etch B (Figure 4b), the HM was excessively consumed at a depth of 5 μm , leaving insufficient protection for the remaining PI layer, preventing completion of the via etch. Therefore, deep via etching requires reducing the fraction of etch B. Concurrently, a low proportion of etch A also caused side grooves, similar to those observed in trench etching. More critically, PI etching residues accumulated on the HM surface under gas flow (Figure 4c). Such accumulation complicates mask removal and may result in static charge buildup that can trigger arcing, potentially damaging wafers and equipment [5-6]. By optimizing the etch gas flow rate and finely adjusting the etch A/B ratio, high-quality deep via etching was achieved, yielding a depth of 7.5 μm , a sidewall angle greater than 85°, and no considerable side grooves (Figure 4d). Sub-5 μm via is currently under development.

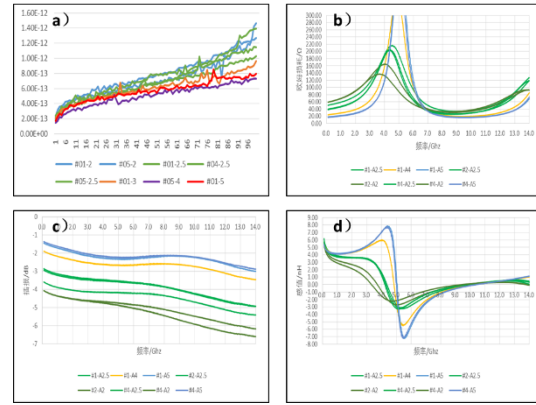


Figure 5. Electrical Performance of Different L/S

- a) Insulation characteristics; b) Impedance characteristics; c) Insertion loss; d) Inductance characteristics.

To characterize the passive properties of the FLLs, test element groups (TEGs) for resistance, inductance, and S-parameters were designed. The electrical performance of wiring with different L/S specifications was measured, and the results are presented in Figure 5. Insulation resistance tests (Figure 5a) depicted that as the line width decreased from 5 μm to 2 μm , insulation resistance decreased from $1.4 \times 10^{14} \Omega$ to $7.7 \times 10^{13} \Omega$. These values remain far above the typical $10^{12} \Omega$ threshold for advanced packaging, demonstrating that fine-lines continue to provide reliable insulation performance. For impedance characteristics (Figure 5b), the DC resistance of 2 μm wiring ($\sim 70 \Omega$) was markedly higher than that of 5 μm wiring ($\sim 20 \Omega$) at 1 GHz, primarily due to the increased resistivity resulting from the reduced conductor cross-sectional area.

High-frequency measurements (Figure 5c) showed that in the 10 GHz band, the insertion loss for 2 μm wiring was approximately 6 dB, compared with 2 dB for 5 μm wiring. This difference is mainly caused by the enhanced skin effect and the potential increase in radiation loss associated with narrower line widths. For inductance characteristics (Figure 5d), the inductance of 2 μm wiring ($\sim 3 \text{ nH}$) was slightly lower than that of 5 μm wiring ($\sim 4 \text{ nH}$), as the reduced line width lowers magnetic flux coupling, which benefits phase stability for high-frequency signals. Overall, the FLLs meets the electrical performance requirements for mid-to high-frequency packaging applications.

Table 3. Reliability Test

Item	Condition	Result
uHAST	130 deg.C/85 %RH 96 hours.	Pass
TCT	-55~125°C, 15min, 1000 cycles	Pass

To meet long-term operational reliability requirements, the reliability of the samples was evaluated, as shown in Table 3. After 96h of uHAST, no failure phenomena such as film peeling, voids, or corrosion, were observed for any L/S specification. After 1000cycles, the resistance variation of the interconnect structures remained within 5% and no degradation in insulation performance was detected. These results indicate that the PI insulating layer and the overall structure possess excellent resistance to moisture/heat aging and thermo-mechanical fatigue, meeting the long-term reliability requirements of advanced packaging products.

4. Conclusion

This study successfully developed a high-density RDL hybrid substrate featuring monolithic integration of an organic interposer with a glass substrate. The core achievements are as follows:

1. An asymmetric structure integrating the FLL directly onto the glass substrate was proposed, eliminating the solder interface between the interposer and the substrate. This approach simplifies the overall packaging process, reduces assembly cost, improves yield, and enables large-form-factor packaging.
2. Through the co-optimization of the embedded RDL process and CMP, key challenges such as hard mask selection and etch profile control were addressed. This enabled the fabrication of high-density interconnects with an L/S of $2/2\ \mu\text{m}$, providing a wiring density 2.5 times greater than that of conventional organic substrates.
3. The limitations associated with traditional laser drilling were overcome through dry etching, achieving high-quality microvias with a $10\text{-}\mu\text{m}$ diameter and a sidewall verticality exceeding 85° , thereby satisfying the needs of high-density three-dimensional interconnect structures.
4. Electrical characterization demonstrated that the $2/2\ \mu\text{m}$ interconnects exhibit high insulation resistance (up to $7.7 \times 10^{13}\ \Omega$) and an insertion loss of 6 dB at 10 GHz, meeting mid- to high-frequency packaging applications. Comprehensive reliability testing confirmed their excellent long-term operational stability, establishing a robust foundation for practical deployment.

5. References

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